ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY AND APPLIED SCIENCES

2023 SESSIONAL EXAMINATIONS

OBJECT ORIENTED PROGRAMMING EXAM (CCS 2121)

INSTRUCTIONS

(1) Time allowed is **Three (3) hours**.

(2) There are Seven (7) questions in this paper. **Answer All Questions in Section A and ANY Two Questions from Section B**

(3) Each question carries **Twenty (20)** marks.

(4) Calculators are not allowed.

(5) The total marks for this Exam is **100**

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SECTION A: MANDATORY QUESTIONS

Question one: Fill in the blanks (20 Marks)

1. In Java, the main method must be declared as _____.
2. The process of combining variables and methods into a single unit is called _____.
3. A variable that is declared inside a method is called a _____ variable.
4. The default value of a boolean variable in Java is _____.
5. To declare a constant variable in Java, you use the _____ keyword.
6. In Java, the keyword used to create an instance of a class is _____.
7. To compare two string objects for equality in Java, you should use the _____ method.
8. The _____ operator is used for concatenating two strings in Java.
9. The _____ class is used to read input from the user in Java.
10. To declare an array in Java, you specify the type of the elements followed by _____.
11. The _____ keyword is used to refer to the current instance of a class.
12. In Java, the _____ statement is used to exit a loop prematurely.
13. The _____ class is used to format and parse date and time in Java.
14. The _____ class provides methods for mathematical operations in Java.
15. The _____ class is the superclass of all exception classes in Java.
16. The _____ statement is used to handle exceptions in Java.
17. The _____ modifier makes a variable or method accessible only within its own class.
18. The _____ method is used to convert a string to an integer in Java.
19. The _____ class provides methods for reading and writing files in Java.
20. The _____ data structure stores elements in a sorted order in Java.

Question two

You have been assigned to develop a Library Management System for a university. The system should keep track of books, library members, and borrowing transactions. The following requirements have been identified and presented to you for implementation:

i. **Books:**

Each book has a unique ISBN, title, author, and availability status (whether it is currently available or not). Books can be added, removed, and searched in the system. The system should display a list of available books.

ii. **Members:**

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Each library member has a unique ID, name, and a list of borrowed books. Members can be added, removed, and searched in the system. The system should display a list of all members and the books they have borrowed.

iii. **Transactions:**

Members can borrow books from the library. Each borrowing transaction should be recorded with the member ID, book ISBN, and the date of borrowing. Members can return books to the library. The system should display a list of all borrowing transactions.

Task: You are required to design and implement a Java program that satisfies the above requirements. **[20 Marks]**

SECTION B: ANSWER ANY THREE QUESTIONS

Question Three

a) Explain the following terminologies as used in Object Oriented Programming **[12 Marks]**

i. Classes
ii. Methods
iii. Objects
iv. Encapsulation
v. Inheritance
vi. Polymorphism

b) Define a constructor and explain why they are important in Java for several reasons. **[2 Mark]**

c) List and explain at least six (6) characteristics of Java programming language. **[6 Marks]**

Question Four

Write a class to demonstrate the use of classes, methods, and objects in a practical program. In this exercise, you are required to define a Student class that has properties such as name, age, and GPA, as well as a method to print the student's information. Then create two Student objects in the Main class and call the printInfo method on each object to print their information. **[20 Marks]**

Question Five

Define a class named Rectangle with private fields length and width, and public methods getLength(), getWidth(), setLength(), setWidth(), and getArea(). Implement the program by

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creating two Rectangle objects, sets their length and width using the setLength() and setWidth() methods, and then prints their areas using the getArea() method. **[20 Marks]**

Question Six

With the use of clear practical examples, provide the syntaxes of the following control structure statements **[20 Marks]:**

i. If statement
ii. If else statement
iii. While Loop
iv. Do while loop
v. For statement
vi. Switch statement

Question Seven

You have been employed as a programmer for a shipping company in Lusaka. You are required to create a Java program that calculates the shipping costs for a package based on its weight. The program should take user input for the weight of the package in kilograms.

To calculate the shipping cost, you need to consider different rates based on weight ranges. For example, if the weight is less than or equal to 5 kg, the cost per kilogram is K 2. If the weight is between 5 kg and 10 kg (inclusive), the cost per kilogram is K 1.5. If the weight is greater than 10 kg, the cost per kilogram is K 1.

a) Use an if-else if statement to determine which weight range the package falls into, and then calculate the total shipping cost accordingly. Once the total shipping cost is calculated, it should be displayed to the user **[10 Marks].**

b) Implement the same program using the switch statement **[10 Marks].**

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